train input, full runs — small-dm

L1D 64 sets x 1, L2 512 sets x 1 · metric schema 9 · gem5 TimingSimpleCPU · 4 November 2026

31 of 32 runs completed, 1 running, 0 pending; **1 remaining**.

Program (does not depend on the cache)

run	configs done	spill/ROI	spill ld/load	spill st/store	(spill+csave)/ROI	(spill+csave) ld/load	(spill+csave) st/store
cactuBSSN	1/1	21.33%	51.32%	55.49%	21.43%	51.50%	55.81%
parest	1/1	0.18%	0.45%	1.28%	0.49%	0.93%	7.38%
deepsjeng	1/1	2.99%	9.41%	15.98%	8.19%	23.80%	48.29%
leela	1/1	0.43%	1.55%	2.04%	6.08%	15.99%	46.72%
namd	1/1	3.22%	8.11%	17.74%	3.23%	8.13%	17.81%
lbm	1/1	9.69%	27.97%	35.99%	9.69%	27.97%	35.99%
nab	1/1	7.61%	25.91%	37.53%	8.29%	27.38%	44.95%
nab_gcn4dna	1/1	7.30%	24.89%	35.20%	8.25%	26.98%	45.65%
xz	1/1	1.36%	4.25%	4.82%	6.14%	16.52%	26.27%
xz_combined	1/1	2.79%	10.48%	25.95%	4.09%	14.43%	41.80%
imagick	1/1	0.04%	0.21%	4.19%	0.25%	0.82%	48.78%
blender	1/1	0.31%	0.55%	4.03%	3.18%	5.93%	39.04%
povray	1/1	0.27%	0.46%	0.95%	14.26%	22.21%	55.39%
omnetpp	1/1	0.01%	0.02%	0.03%	12.59%	24.64%	48.38%
xalancbmk	1/1	0.44%	1.00%	1.04%	18.29%	32.42%	63.05%
x264	1/1	6.30%	13.42%	40.78%	8.75%	18.41%	57.31%
gcc_200	1/1	1.16%	2.64%	4.34%	16.68%	35.31%	67.49%
gcc_scilab	1/1	1.25%	2.90%	4.70%	15.74%	33.65%	65.08%
gcc	1/1	1.41%	3.09%	5.21%	16.29%	32.84%	65.64%
mcf	1/1	1.61%	2.27%	13.28%	2.75%	4.04%	21.88%
perlbench_scrabbl	1/1	0.23%	0.30%	0.93%	17.65%	32.90%	56.14%
perlbench_suns	1/1	0.98%	2.06%	2.86%	18.85%	36.45%	59.85%
perlbench_diffmail	1/1	2.06%	4.51%	4.76%	17.54%	31.15%	53.97%
perlbench_perfect	1/1	1.44%	2.34%	5.00%	18.85%	33.31%	60.43%
perlbench_splitmail	1/1	5.79%	9.36%	20.68%	17.56%	30.55%	58.86%
bwaves_1	1/1	2.99%	7.65%	18.86%	2.99%	7.65%	18.86%
bwaves_2	1/1	3.04%	7.82%	18.83%	3.04%	7.82%	18.83%
exchange2	1/1	4.84%	10.67%	12.41%	6.03%	13.09%	15.72%
fotonik3d	1/1	3.99%	10.66%	12.65%	4.00%	10.69%	12.75%
roms	0/1	—	—	—	—	—	—
wrf	1/1	0.92%	3.44%	2.61%	1.59%	4.97%	8.09%
cam4	1/1	2.49%	9.15%	7.96%	4.59%	14.65%	19.40%
pooled		3.60%	9.44%	17.30%	7.34%	17.15%	41.91%
mean		3.18%	8.35%	13.49%	9.59%	20.87%	41.47%

Miss rates

run	state	L1D	spill ld L1	other ld L1	spill st L1	other st L1	L2	spill ld L2	spill st L2
cactuBSSN	done	17.36%	6.42%	36.79%	5.50%	16.60%	48.81%	11.44%	8.83%
parest	done	15.29%	13.83%	16.21%	5.54%	3.80%	50.52%	24.84%	9.30%
deepsjeng	done	13.57%	6.64%	17.09%	9.69%	7.53%	20.58%	4.92%	19.04%

run	state	L1D	spill ld L1	other ld L1	spill st L1	other st L1	L2	spill ld L2	spill st L2
leela	done	7.77%	12.09%	8.63%	0.23%	5.06%	15.93%	0.08%	54.80%
namd	done	15.53%	11.95%	18.84%	2.50%	3.73%	37.27%	8.80%	21.24%
lbm	done	26.83%	1.63%	13.80%	5.34%	81.66%	74.99%	5.20%	5.54%
nab	done	11.61%	4.17%	16.71%	3.10%	1.82%	16.55%	1.10%	1.00%
nab_gcn4dna	done	11.34%	4.33%	16.03%	3.99%	1.54%	31.28%	3.87%	1.80%
xz	done	8.85%	3.59%	11.81%	9.75%	4.03%	32.06%	1.52%	2.69%
xz_combined	done	14.68%	7.09%	17.42%	5.23%	8.97%	25.02%	1.95%	1.77%
imagick	done	5.62%	28.95%	5.61%	2.48%	3.03%	86.53%	17.31%	2.92%
blender	done	9.96%	6.67%	10.51%	7.11%	6.45%	27.63%	13.01%	14.24%
povray	done	14.30%	12.02%	16.75%	21.55%	8.08%	36.10%	32.70%	49.08%
omnetpp	done	18.54%	0.72%	24.19%	10.69%	7.45%	41.67%	7.36%	0.74%
xalancbmk	done	14.58%	30.23%	18.61%	25.47%	6.18%	26.71%	27.23%	43.28%
x264	done	4.04%	0.93%	4.85%	1.20%	4.46%	26.31%	3.81%	16.05%
gcc_200	done	12.95%	7.30%	16.90%	8.80%	5.66%	37.34%	3.02%	9.41%
gcc_scilab	done	13.63%	7.84%	17.82%	8.20%	5.88%	38.00%	4.27%	10.91%
gcc	done	13.47%	6.70%	17.48%	8.24%	5.86%	39.11%	2.99%	9.24%
mcf	done	26.68%	10.49%	30.85%	10.54%	8.39%	71.66%	9.97%	20.62%
perlbench_scrabbl	done	12.39%	5.42%	15.67%	15.50%	6.79%	16.92%	0.00%	1.00%
perlbench_suns	done	9.86%	2.95%	12.95%	10.61%	4.92%	25.17%	2.61%	13.53%
perlbench_diffmail	done	10.33%	2.39%	13.31%	10.54%	5.51%	26.66%	0.94%	10.19%
perlbench_perfect	done	11.06%	1.54%	14.81%	7.69%	4.77%	49.94%	19.62%	12.13%
perlbench_splitmail	done	8.89%	3.08%	12.28%	2.97%	4.68%	21.52%	0.16%	7.66%
bwaves_1	done	43.95%	6.18%	48.39%	1.30%	42.07%	26.39%	19.27%	1.31%
bwaves_2	done	43.91%	6.08%	48.40%	1.32%	42.55%	26.65%	19.10%	1.31%
exchange2	done	3.65%	2.78%	4.62%	2.44%	2.62%	1.48%	0.21%	0.90%
fotonik3d	done	22.47%	9.88%	28.82%	0.43%	6.62%	36.82%	11.33%	2.27%
roms	running	—	—	—	—	—	—	—	—
wrf	done	25.82%	13.72%	27.42%	8.46%	22.13%	26.35%	14.83%	15.87%
cam4	done	14.71%	5.03%	15.94%	3.66%	15.14%	46.92%	11.55%	24.53%
pooled		14.84%	5.98%	17.76%	4.80%	9.95%	36.07%	10.04%	11.39%
mean		15.60%	7.83%	18.69%	7.10%	11.42%	35.13%	9.19%	12.68%

Formulas

column	formula	aciklama
spill/ROI	`((spill_load + spill_store) / roi_total_insts) * 100`	ROI komutlarinin yuzde kaci spill load veya spill store
spill ld/load	`(spill_load / roi_load) * 100`	ROI load'larinin yuzde kaci spill reload
spill st/store	`(spill_store / roi_store) * 100`	ROI store'larinin yuzde kaci spill store
(spill+csave)/ROI	`((spill_load + spill_store + csave_load + csave_store) / roi_total_insts) * 100`	ayni oran, callee-save save/restore dahil
(spill+csave) ld/load	`((spill_load + csave_load) / roi_load) * 100`	ROI load'larinin yuzde kaci spill reload veya callee-save restore
(spill+csave) st/store	`((spill_store + csave_store) / roi_store) * 100`	ROI store'larinin yuzde kaci spill store veya callee-save save
L1D	`(L1D misses / L1D accesses) * 100`	Tum load ve store access'lerinin (spill + other) L1D miss rate'i: L1D'nin bu konfigurasyondaki genel performansi.
spill ld L1	`(spill load L1D misses / spill load L1D accesses) * 100`	Spill load'larin yuzde kaci L1D'de miss oldu. Genelde dusuktur: spill slot'u kisa sure once yazildigi icin cache line cogu zaman hala L1D'dedir.
other ld L1	`((load misses - spill load misses) / (load accesses - spill load accesses)) * 100`	Other load'larin (spill olmayan load'lar) L1D miss rate'i. spill_ld_L1miss ile yan yana bakilir: spill load'lar normal veriden daha mi az, daha mi cok miss oluyor?

column	formula	aciklama
spill st L1	`(spill store L1D misses / spill store L1D accesses) * 100`	Spill store'larin yuzde kaci L1D'de miss oldu. Cache write-allocate: miss olan bir store once cache line'i L2'den getirir, sonra yazar.
other st L1	`((store misses - spill store misses) / (store accesses - spill store accesses)) * 100`	Other store'larin (spill olmayan store'lar) L1D miss rate'i; spill_st_L1miss ile karsilastirmak icin.
L2	`(L2 misses / L2 accesses) * 100`	L2'ye gelen tum isteklerin (L1D ve L1I miss'leri) yuzde kaci L2'de de miss oldu ve DRAM'e gitti.
spill ld L2	`(spill load L2 misses / spill load L2 accesses) * 100`	L1D'de miss olan spill load'larin yuzde kaci L2'de de miss oldu. Payda sadece L1D'yi miss eden spill load'lardir, butun spill load'lar degil.
spill st L2	`(spill store L2 misses / spill store L2 accesses) * 100`	L1D'de miss olan spill store'larin yuzde kaci L2'de de miss oldu (payda yine sadece L1D miss'leri).

spill = compiler-confirmed register-allocator spill (GCC `-dP`), csave = callee-saved register save/restore (CFI), both inside the ROI only. pooled = sum of numerators / sum of denominators over the completed runs; mean = average of the per-run percentages. Miss rates are per cache access; an L2 rate's denominator is the L1D misses of that access type.